

1 Ans: B

The term $\frac{1}{e^{\left(\frac{hf}{kT}\right)} - 1}$ is unitless.

Hence comparing units: $[(D^3)] = \frac{[R] \times [c^2]}{[h]}$ where $[]$ represents “units of”.

Since units of R are $\text{J s}^{-1} \text{m}^{-2}/\text{s}^{-1}$, units of c^2 are $\text{m}^2 \text{s}^{-2}$ and units of h are J s , units of D^3 are s^{-3} . Hence D has unit s^{-1} .

2 Ans: B